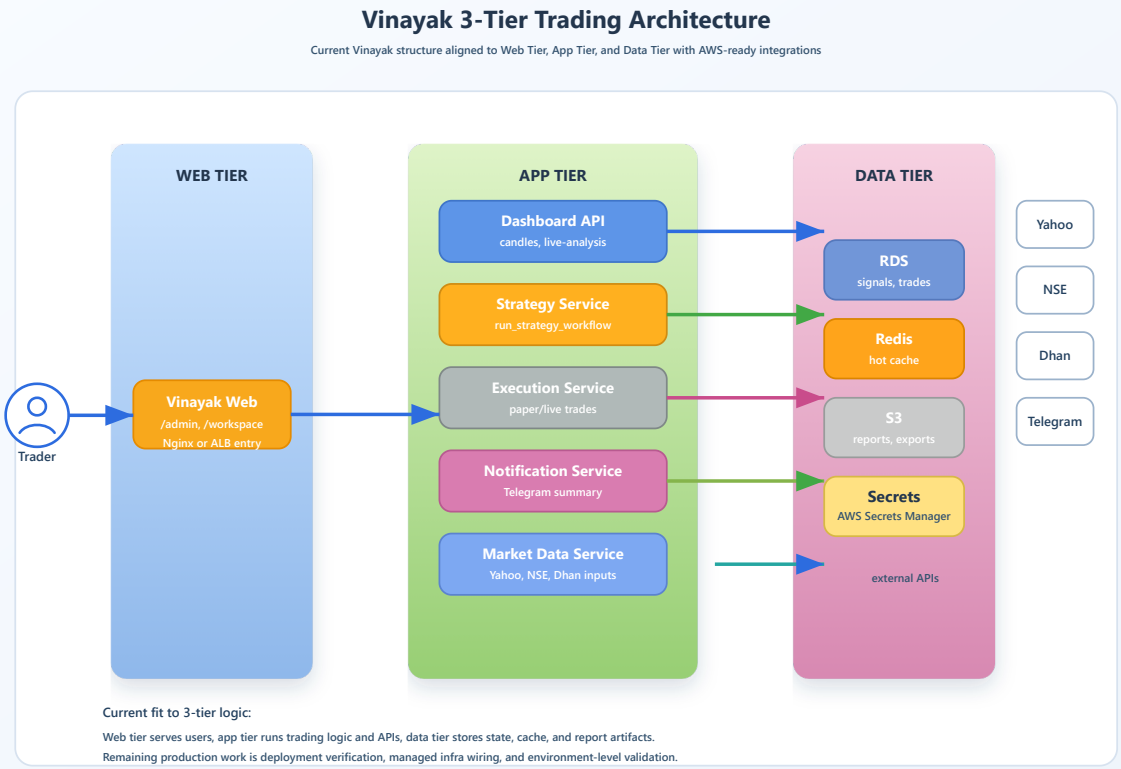
Vinayak 3-Tier Project Pack

Professional combined document pack for the Vinayak trading platform covering architecture, implementation, and production rollout flow.

Architecture Diagram | Web Tier | App Tier | Data Tier | Implementation Flow

Architecture Summary

Vinayak now behaves like a 3-tier trading platform in structure. Browser traffic is handled by the web tier, business logic runs in the app tier, and persistence plus cache plus report artifacts belong to the data tier.



Web Tier

- Admin and workspace pages

App Tier

- FastAPI routes and services

- Nginx or ALB entry point
- Public-facing only at ingress layer

- Live analysis, strategy, execution, Telegram
- Legacy Trading logic bridged into Vinayak

Data Tier

- DB-backed trading state
- Redis hot-cache for candles and artifacts
- S3 or local report storage

Production Target

- ALB + ECS Fargate
- RDS + ElastiCache + S3
- Secrets Manager + CloudWatch

Tier Implementation Steps

01 Web Tier

Install Nginx, configure reverse proxy, and expose only the web entry layer.

```
sudo apt update
sudo apt install nginx -y
sudo systemctl enable nginx
sudo systemctl start nginx
```

02 App Tier

Install the Vinayak dependencies and run the FastAPI application.

```
py -3 -m pip install -r vinayak/requirements.txt
py -3 -m uvicorn vinayak.api.main:app --host 0.0.0.0 --port 8000
```

03 Data Tier

Configure database, Redis, and report artifact storage.

```
REDIS_URL=redis://localhost:6379/0
REPORTS_DIR=vinayak/data/reports
REPORTS_S3_BUCKET=your-report-bucket
AWS_REGION=ap-south-1
```

Ordered Rollout

Step 1: Prepare compute and private networking.

Step 2: Put Nginx or ALB in front as the web tier.

Step 3: Start the Vinayak FastAPI service as the app tier.

Step 4: Connect PostgreSQL, Redis, and S3 as the data tier.

Step 5: Add deployment controls, logs, alerts, and CI/CD.

Step 6: Validate end-to-end with live analysis and report artifacts.